

Marcellus E. Parker

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Education

University of Michigan school of Engineering

B.S.E AEROSPACE ENGINEERING

Morehouse College

B.S APPLIED PHYSICS

Illinois Institute of Technology

M.S.E MECHANICAL/AEROSPACE ENGINEERING (COURSEWORK COMPLETED)

Ann Arbor, MI

Mar. 2010 - PRESENT

Atlanta, GA

Mar. 2007 - 2014

Chicago, IL

Mar. 2017 - 2019

Skills

Technical Tools Microsoft Word, Excel, PowerPoint, Project

Programming Python, MATLAB, LaTeX

Engineering Tools Siemens NX, SolidWorks, AutoCAD, MagNet, Abaqus, ANSYS

Engineering Domains High Vacuum Systems, Radiation Environments, Thermal Systems, Fluid Systems,

Testing & Validation High-Voltage Electrical Testing, Structural Testing, Failure mode Testing

Experience

SLAC National Laboratory

Palo Alto, CA

STAFF ENGINEER II (ENGINEERING PHYSICIST II)

April 2022 - Present

- Led the restoration and recommissioning of the FACET positron beamline (inactive since 2016), returning RF, magnet, vacuum, and water cooling systems to operational status.
- Directed cross-disciplinary engineering efforts across vacuum, electrical, cooling, and beam diagnostic systems to re-establish beam-line functionality and ensure safe, reliable operation.
- Led the recommissioning of the bunch length diagnostic system after six years offline, restoring a critical beam diagnostic and enabling reliable, high-resolution measurements for accelerator operations and research.
- Developed MATLAB-based tools to process and visualize beam diagnostics data across accelerator sectors, including magnet mover controls and slice emittance and energy spread analysis, improving operational diagnostics and analytical capability for accelerator studies.
- Supported FACET-II accelerator operations and troubleshooting efforts, diagnosing issues across RF, magnet, vacuum, and diagnostic systems to maintain stable beam delivery and operational reliability.
- Engineered electrical configurations to enforce robust Lockout-Tagout (LOTO) conditions, significantly improving equipment safety, maintenance accessibility, and operational reliability.

Fermilab

Batavia, IL

ENGINEERING PHYSICIST I, US-HL-AUP SUPERCONDUCTING R&D ENGINEER & ELECTRICAL QUALITY

April 2015 - March 2022

ASSURANCE

- Held primary responsibility for electrical quality control of Nb₃Sn superconducting magnet systems in the US HiLumi-LHC AUP, improving fabrication reliability and reducing early-stage production defects.
- Performed electromechanical failure analyses on prototype and production Nb₃Sn coils to diagnose quench mechanisms and inform time-critical production decisions.
- Designed and fabricated small-scale R&D superconducting test coils to reproduce quench failure modes observed in pre-production magnet coils, supporting electromechanical failure investigations.
- Designed and implemented a nitrogen (N₂) distribution system for furnace operations to control oxygen during Nb₃Sn coil processing, improving process reliability and reducing technician intervention.
- Developed a MATLAB GUI to analyze thermocouple temperature histories against heat treatment recipes, identifying process control discrepancies and supporting furnace performance validation.
- Managed daily fabrication workflow for Nb₃Sn superconducting magnet coils, supporting pre-production (24 units) and full production (50 units) while coordinating manufacturing priorities to meet delivery schedules.
- Collaborated with physicists and engineers to optimize Nb₃Sn coil heat treatment processes and achieve targeted superconducting performance metrics based on laboratory testing and analysis.
- Executed final electrical qualification testing of AUP superconducting coils prior to shipment to CERN, verifying compliance with specifications and readiness for system integration.

Fermilab

Batavia, IL

ACCELERATOR SYSTEMS & OPERATIONS

Feb 2013 - April 2014

- Coordinated accelerator operations to initiate, optimize, and maintain beamline performance in support of laboratory and university-based research programs.
- Operated, tuned, and maintained the H⁻ plasma source to ensure stable beam delivery across the accelerator complex.
- Contributed to the design and implementation of an air diversion system for the NuMI horn power stripline, improving thermal management and operational reliability for high-power beam delivery.

Georgia Institute of Technology

Atlanta, GA

GRADUATE RESEARCH ASSISTANT

Dec 2013 - April 2014

- Participated in an in-depth study to understand plasma boundary interactions and wall effects.
- Studied plume divergence behavior of the P5 5 kW Hall thruster.
- Assisted in the design of a glass injection tube for a helicon ion thruster.

- Conducted aeroelasticity research on highly flexible aircraft, supporting design and testing of UAV systems.
- Designed and validated a ventral fin using CFD and wind tunnel testing (2'x2' and 5'x7') to improve yaw stability and control authority.
- Developed a sensor-based system using gyroscopes to model and measure aircraft structural deformation in flight.
- Fabricated and maintained structural components for the X-HALE UAV, supporting iterative design improvements.
- Contributed to full-scale assembly of an Aerial Test Vehicle (ATV), including composite wing layup and manufacturing.

Publications & Technical Contributions

- G. Ambrosio *et al.* (including M. Parker), "Lessons Learned from the MQXFA Low-Beta Quadrupoles...", *IEEE Trans. Appl. Supercond.*, 2021.
- V. Marinozzi *et al.*, "Heater-Coil Electrical Insulation for HL-LHC Low Beta Quadrupoles," *IEEE Trans. Appl. Supercond.*, 2021. DOI: 10.1109/TASC.2021.3061348.
- G. Ambrosio *et al.*, "MQXFA Final Design Report," arXiv:2203.06723, 2022.
- C. Clarke *et al.*, "Facet-II," in *Proc. LINAC2022, JACoW*, 2022.
- D. Storey *et al.*, "FACET-II Status and First Results," in *Proc. IPAC2023, JACoW*, 2023.